M|*A*> = |*B*>, and *z* is any complex number, then

$$\mathbf{M}z|A\rangle = z|B\rangle.$$

The only other rule is that, when **M** acts on a sum of vectors, the results are simply added together:

$$\mathbf{M}\{|A\rangle + |B\rangle\} = \mathbf{M}|A\rangle + \mathbf{M}|B\rangle.$$

To give a concrete representation of linear operators, we return to the row and column vector representation of bra- and ket-vectors that we used in Lecture 1. The row-column notation depends on our choice of basis vectors. If the vector space is *N*-dimensional, we choose a set of *N* orthonormal (orthogonal and normalized) ket-vectors. Let's label them |*j*>, and their dual bra-vectors <*j*|.

We are now going to take the equation

$$\mathbf{M}|A\rangle = |B\rangle$$

and write it in component form. As we did in Eq. 1.3, we'll represent an arbitrary ket |*A*> as a sum over basis vectors:

$$|A\rangle = \sum_j \alpha_j |j\rangle.$$

Here, we're using *j* as an index rather than *i* so you won't be tempted to think that we're talking about the *in* spin state. Now, we'll represent |*B*> in the same way and plug both of these substitutions into M|*A*> = |*B*>. That gives